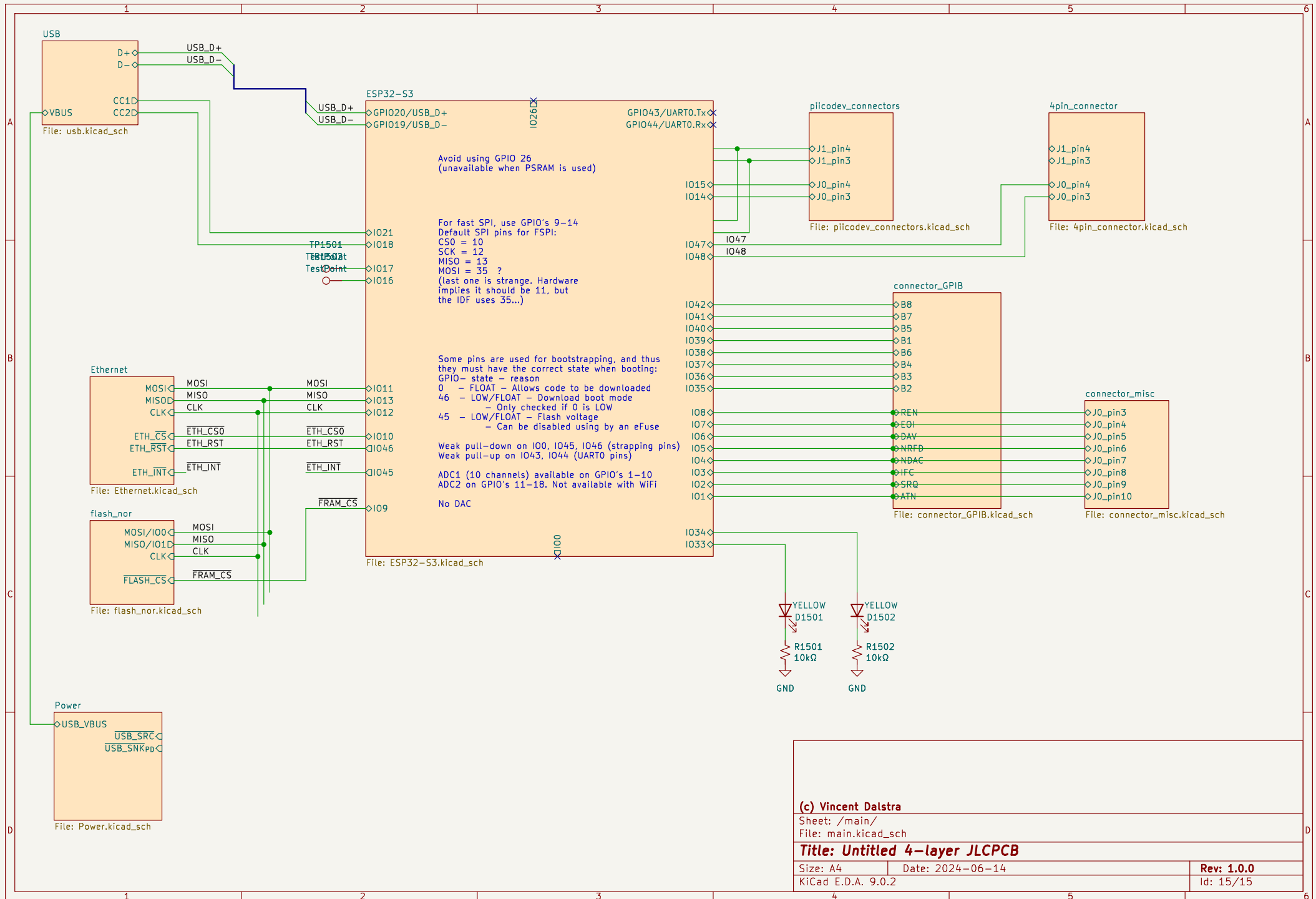


Untitled 4-layer JLCPCB

1	2	3	4	5	6
A	main File: subsheet/main.kicad_sch				
B					
C					
D	Check by: Drawn by: VGD (c) Vincent Dalstra Sheet: / File: ESP_bare_chip_test_board.kicad_sch Title: Untitled 4-layer JLCPCB Size: A4 Date: 2024-06-14 Rev: 1.0.0 KiCad E.D.A. 9.0.2 Id: 1/15				
1	2	3	4	5	6

?? USB alt mode = 2-10
?? Piicodex connector = 2 each
?? 4pin JST connector = 2 each

Untitled 4-layer JLCPCB



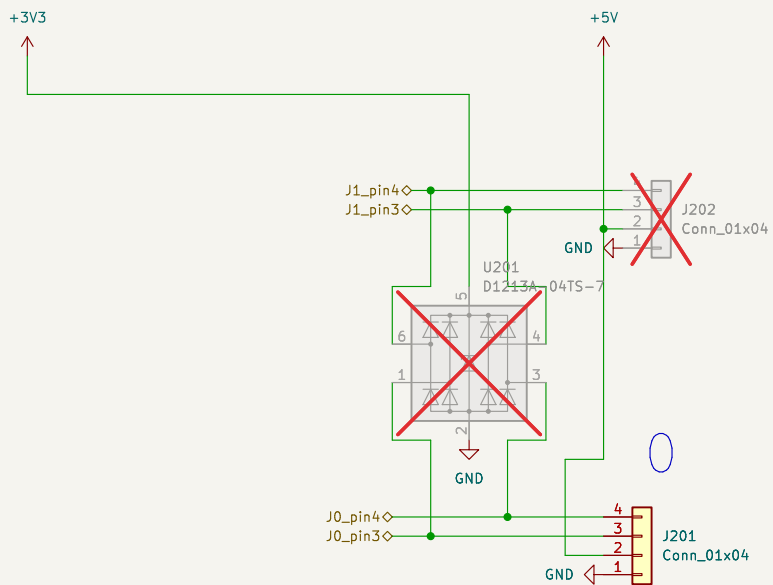
(c) Vincent Dalstra

Sheet: /main/
File: main.kicad_sch

Title: Untitled 4-layer JLCPCB

Size: A4 Date: 2024-06-14
KiCad E.D.A. 9.0.2

Rev: 1.0.0
Id: 15/15



(c) Vincent Dalstra

Sheet: /main/4pin_connector/
File: 4pin_connector.kicad_sch

Title: Untitled 4-layer JLCPCB

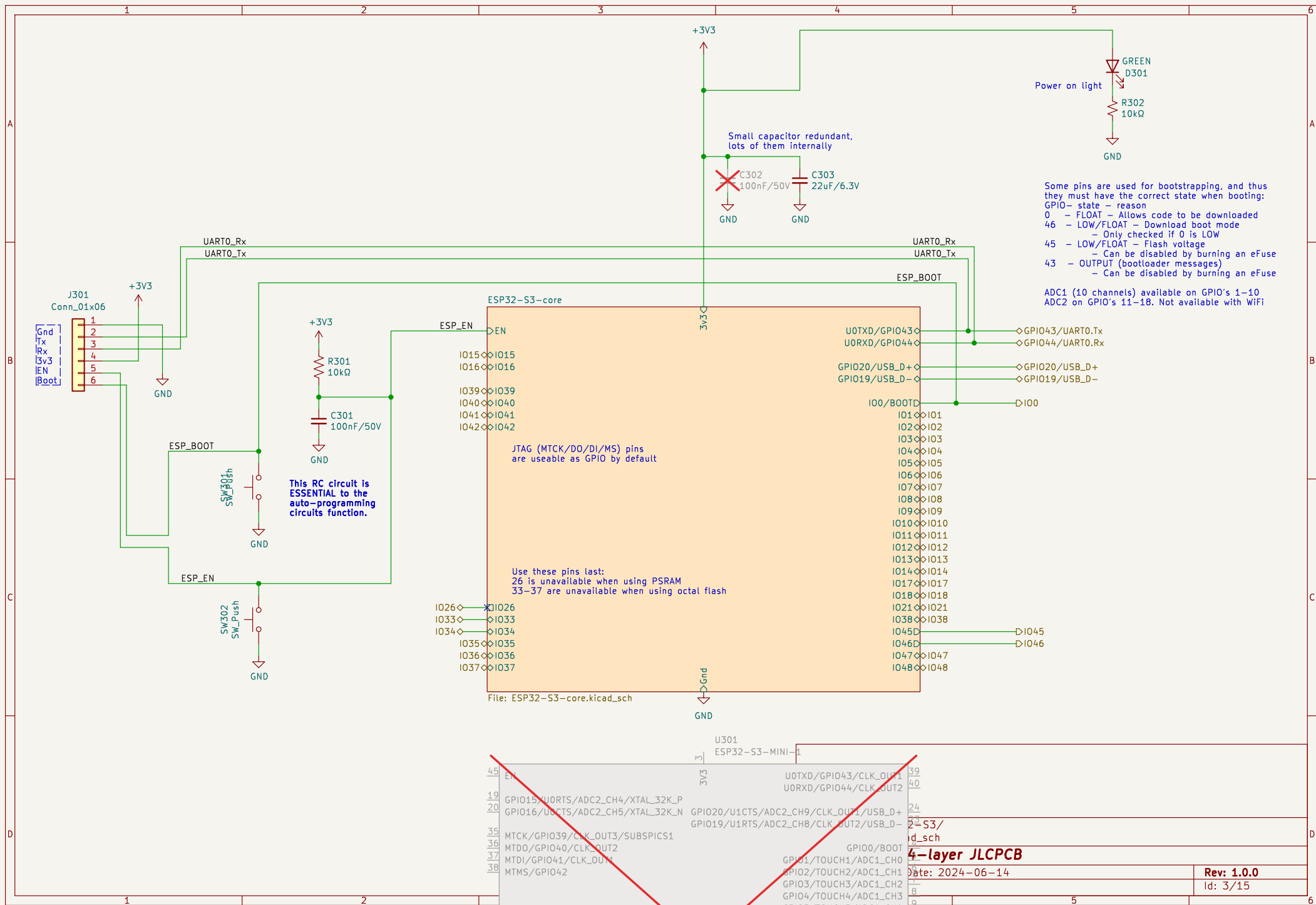
Size: A4

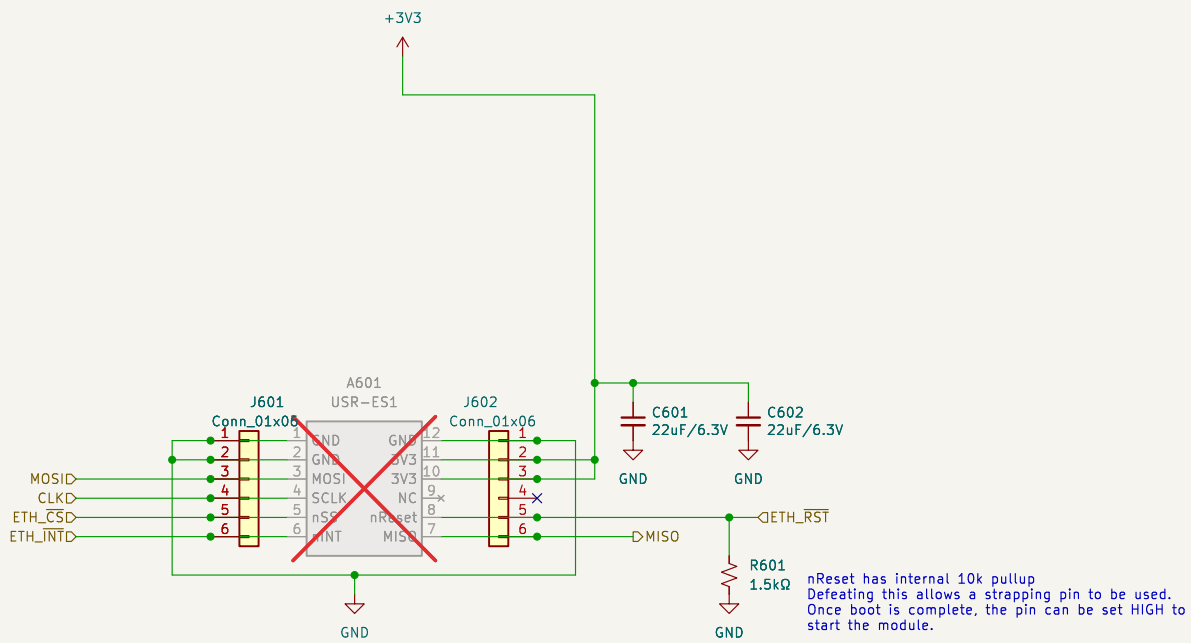
Date: 2024-06-14

Rev: 1.0.0

KiCad E.D.A. 9.0.2

Id: 2/15





Check by:
Drawn by: VGD

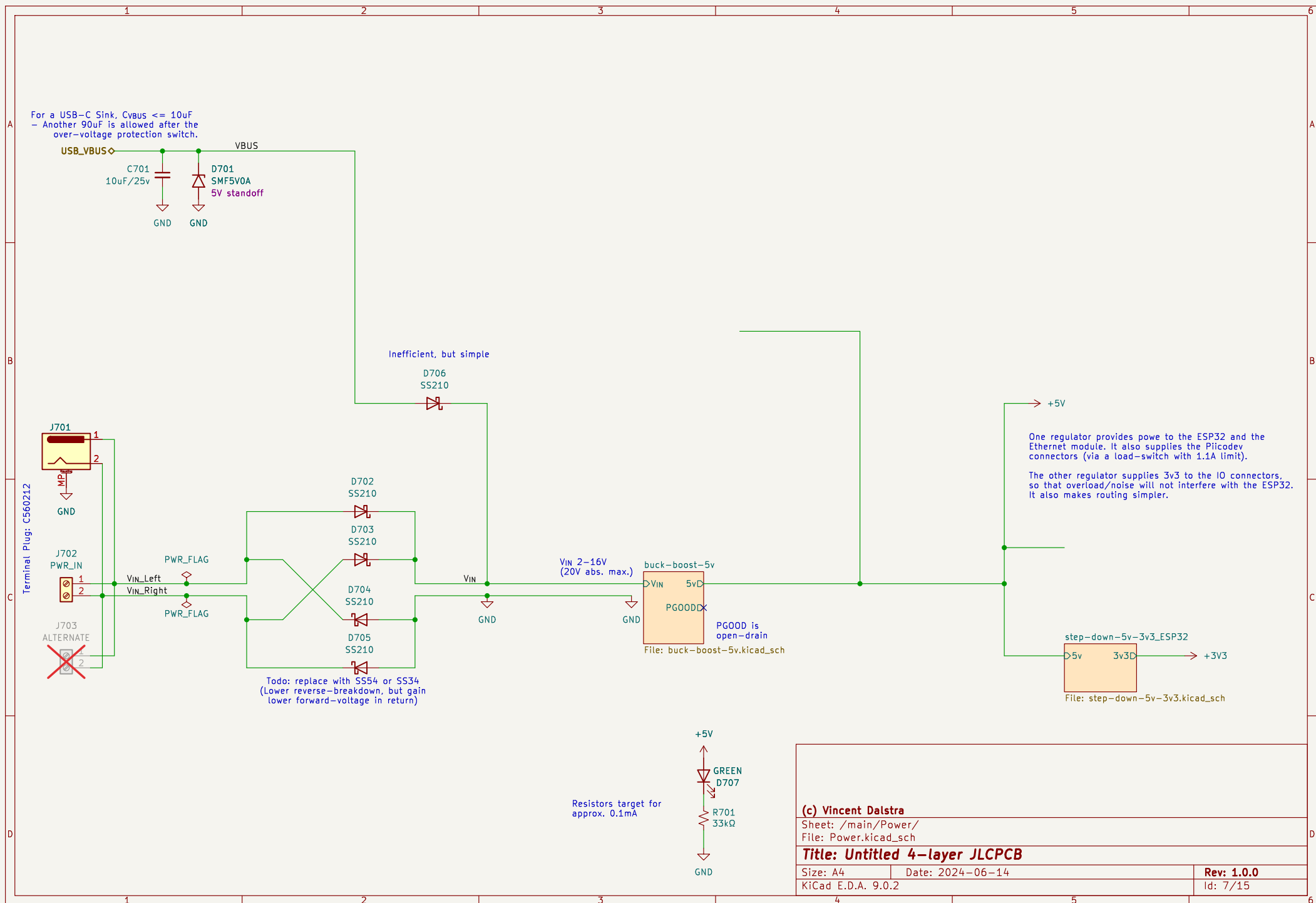
(c) Vincent Dalstra

Sheet: /main/Ethernet/
File: Ethernet.kicad_sch

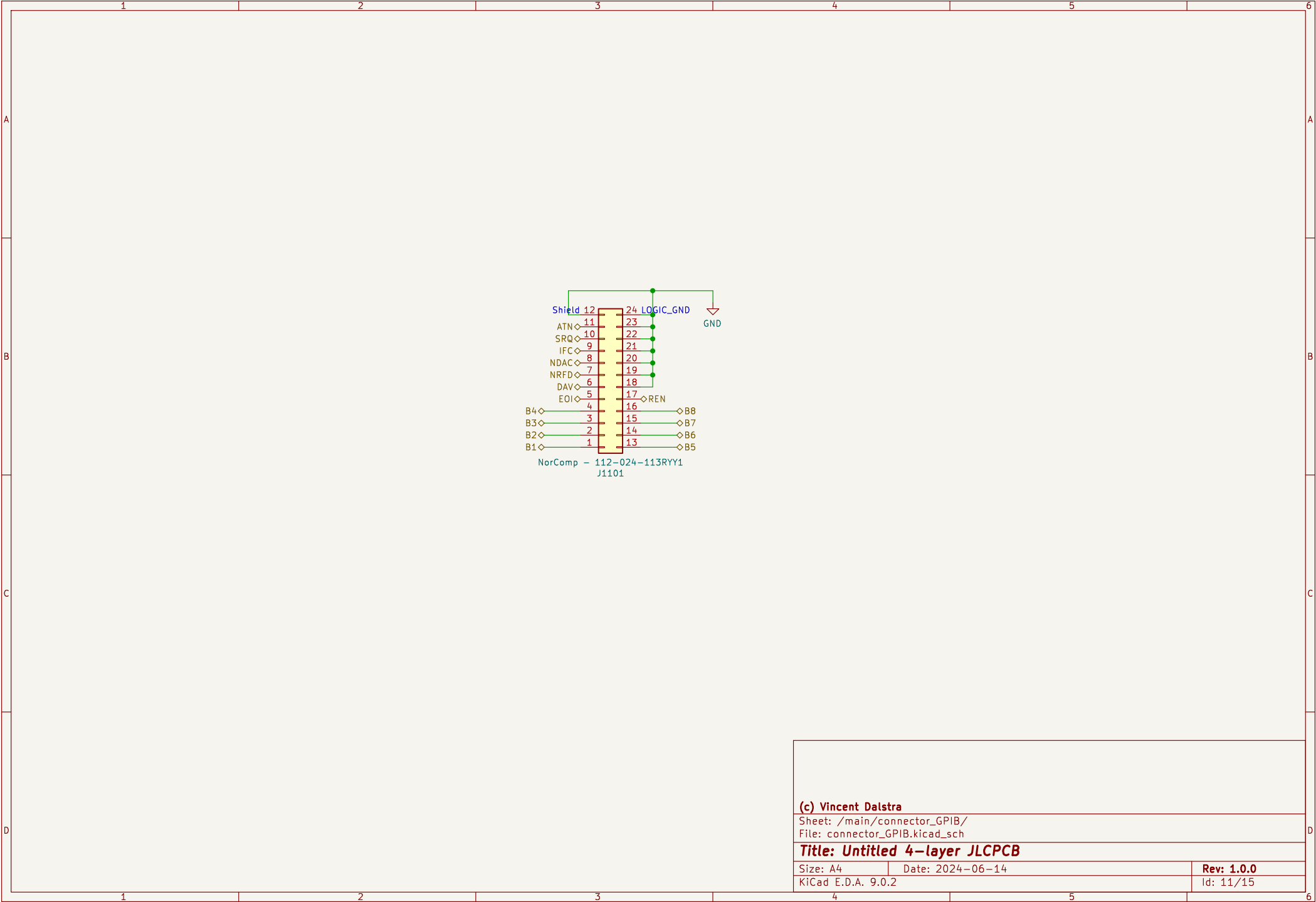
Title: Untitled 4-layer JLCPCB

Size: A4	Date: 2024-06-14	Rev: 1.0.0
KiCad E.D.A. 9.0.2		Id: 6/15

Also provides ESD/surge protection (up to 120V)
OTG functionality allows 0V drop in reverse (like a P-mosfet)
- First the internal body diode conducts to 2.5V
- Then the mosfet is driven high to remove the voltage drop



Id: 10/15



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Sheet: /main/connector_GPIB/
File: connector_GPIB.kicad_sch

Title: Untitled 4-layer JLCPCB

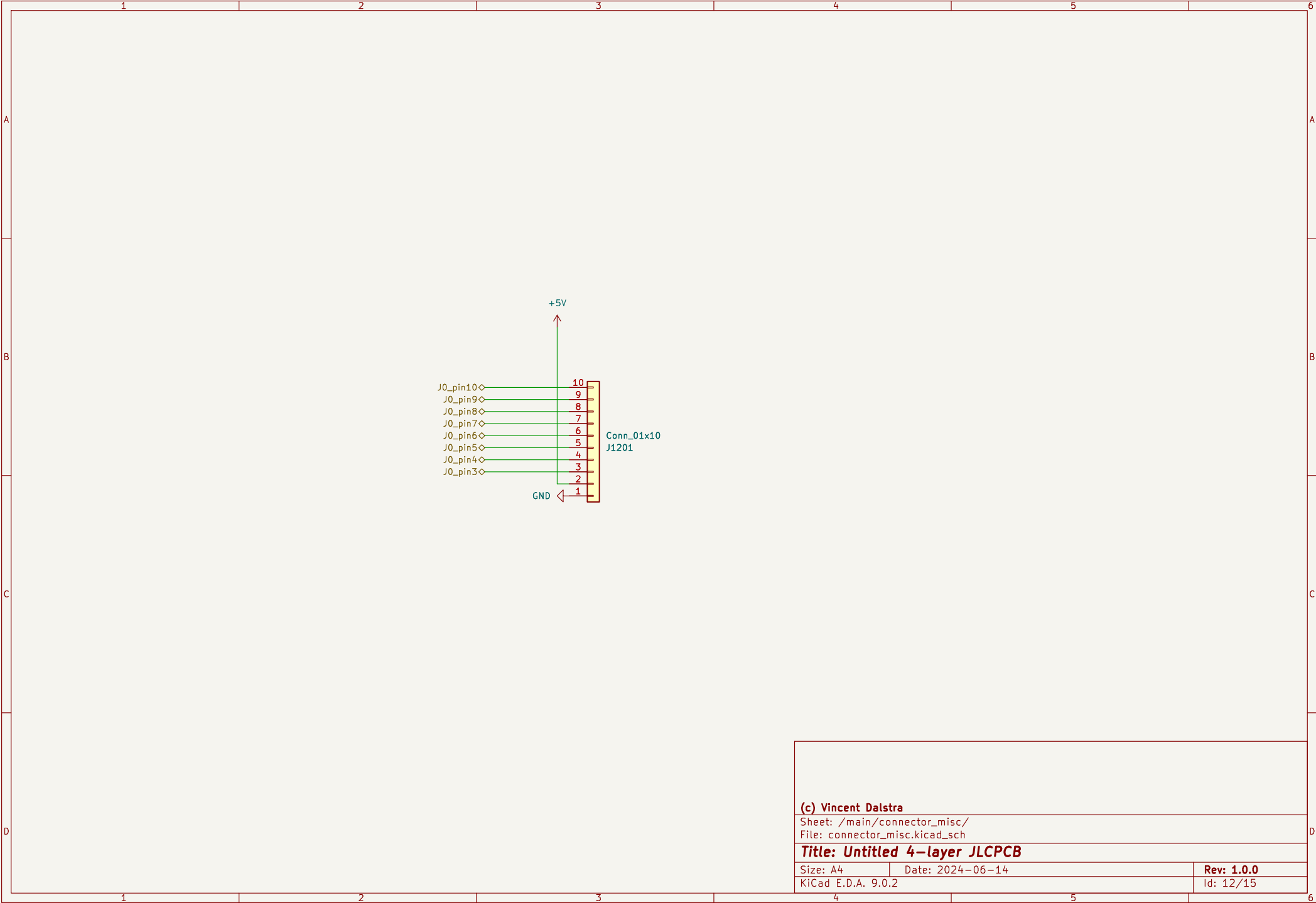
Size: A4

Date: 2024-06-14

Rev: 1.0.0

KiCad E.D.A. 9.0.2

Id: 11/15



(c) Vincent Dalstra

Sheet: /main/connector_misc/
File: connector_misc.kicad_sch

Title: Untitled 4-layer JLCPCB

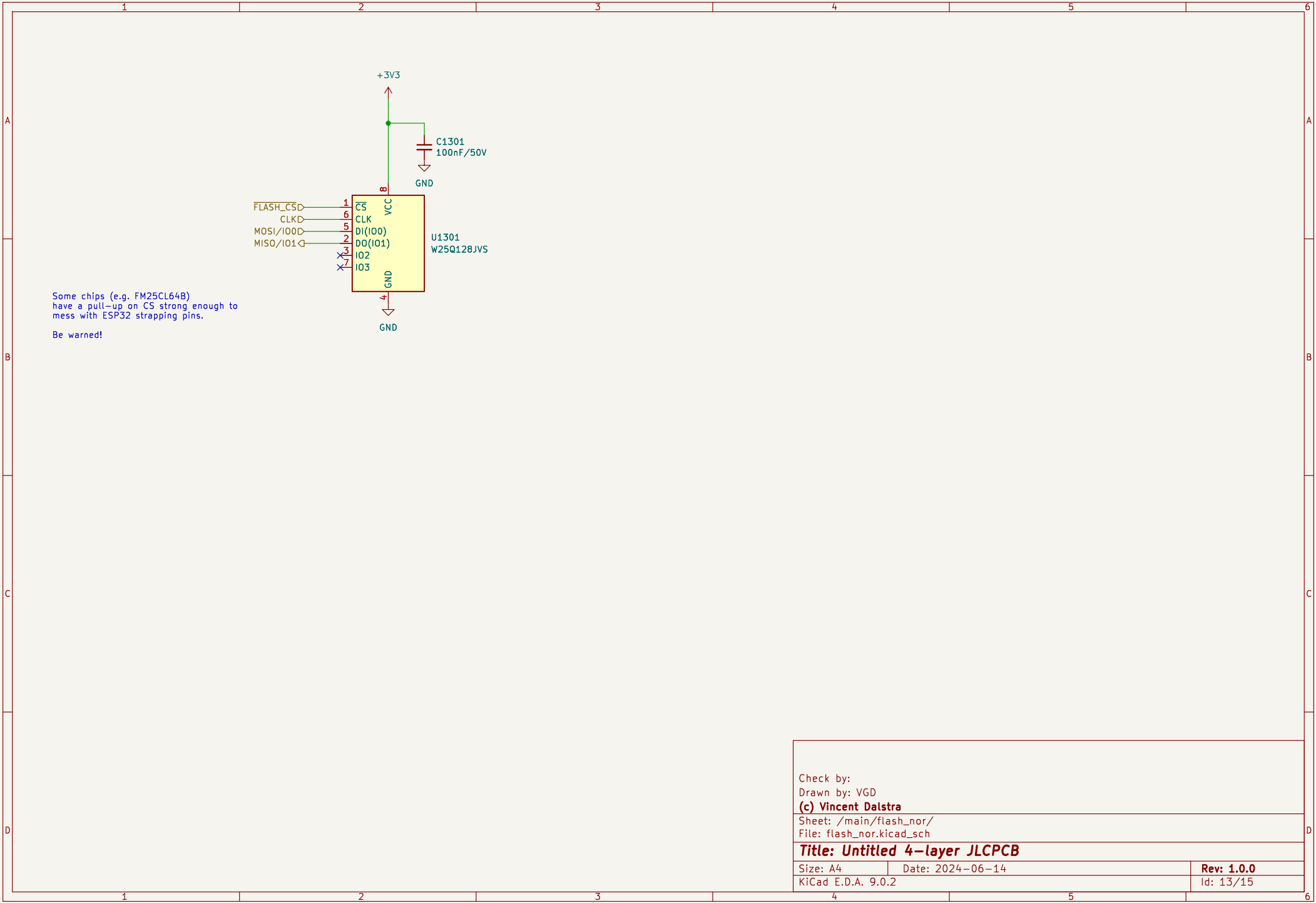
Size: A4

Date: 2024-06-14

Rev: 1.0.0

KiCad E.D.A. 9.0.2

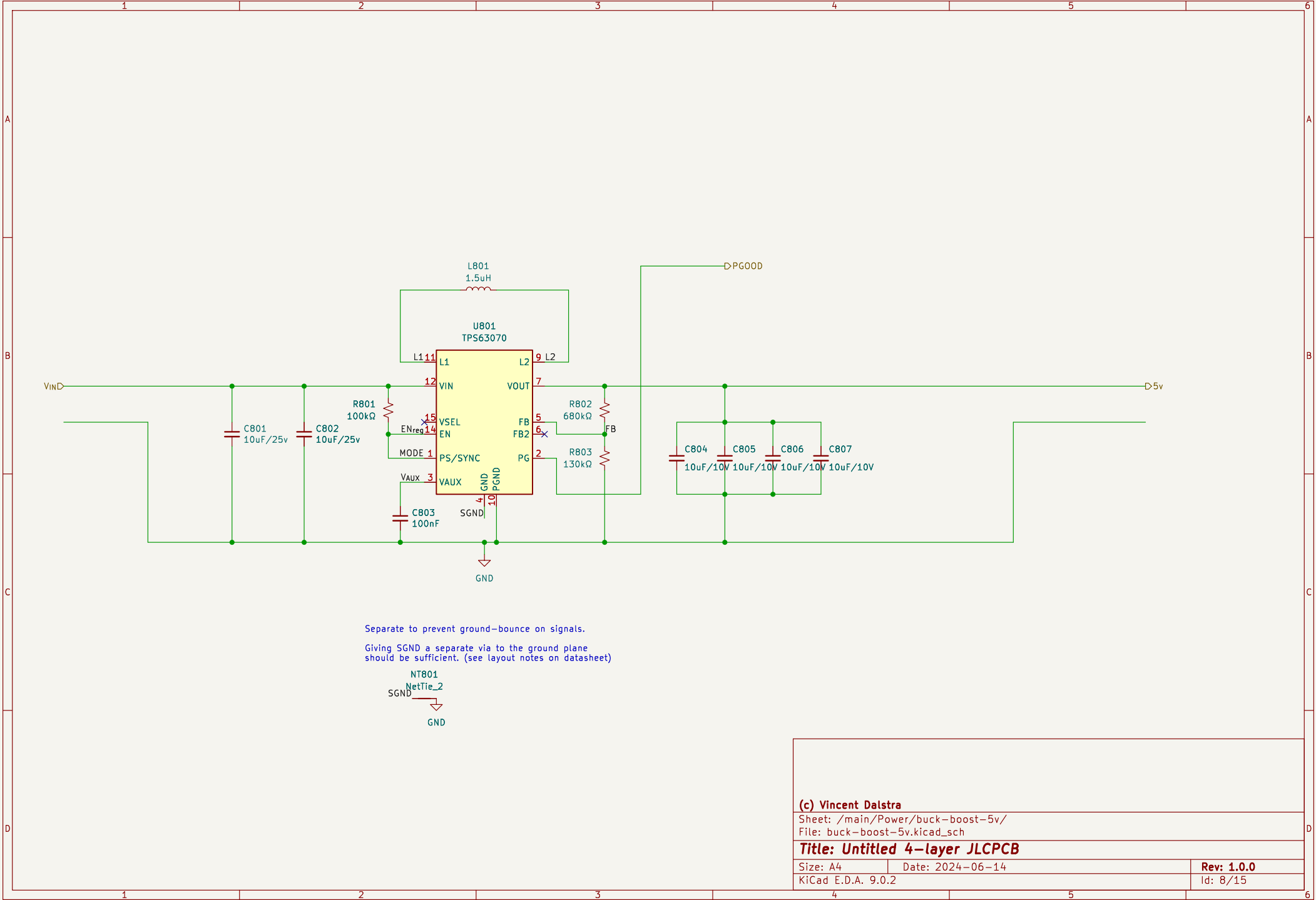
Id: 12/15





Id: 14/15



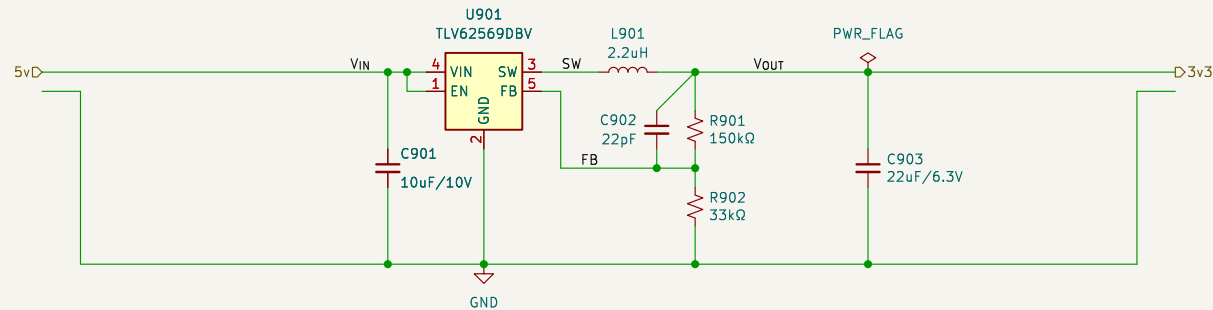


Using WeBench circuit designer
3.3v 2A output (same results for 1A)
"Balanced"

Added Feed-forward capacitor as well

When selecting inductor, consider
max current AND saturation current

$$\begin{aligned} I_{PEAK} &= I_{MAX} + (I_{RIPPLE} / 2) \\ &= 2A + 0.5 * 0.33A \\ &= 2.2A \end{aligned}$$



Feed-forward capacitor
improves transient response

Recommened value = 6.8pF
for Rbottom = 100k
(Inversely proportional)

see datasheet for details.

Check by:
Drawn by: VGD

(c) Vincent Dalstra

Sheet: /main/Power/step-down-5v-3v3_ESP32/
File: step-down-5v-3v3.kicad_sch

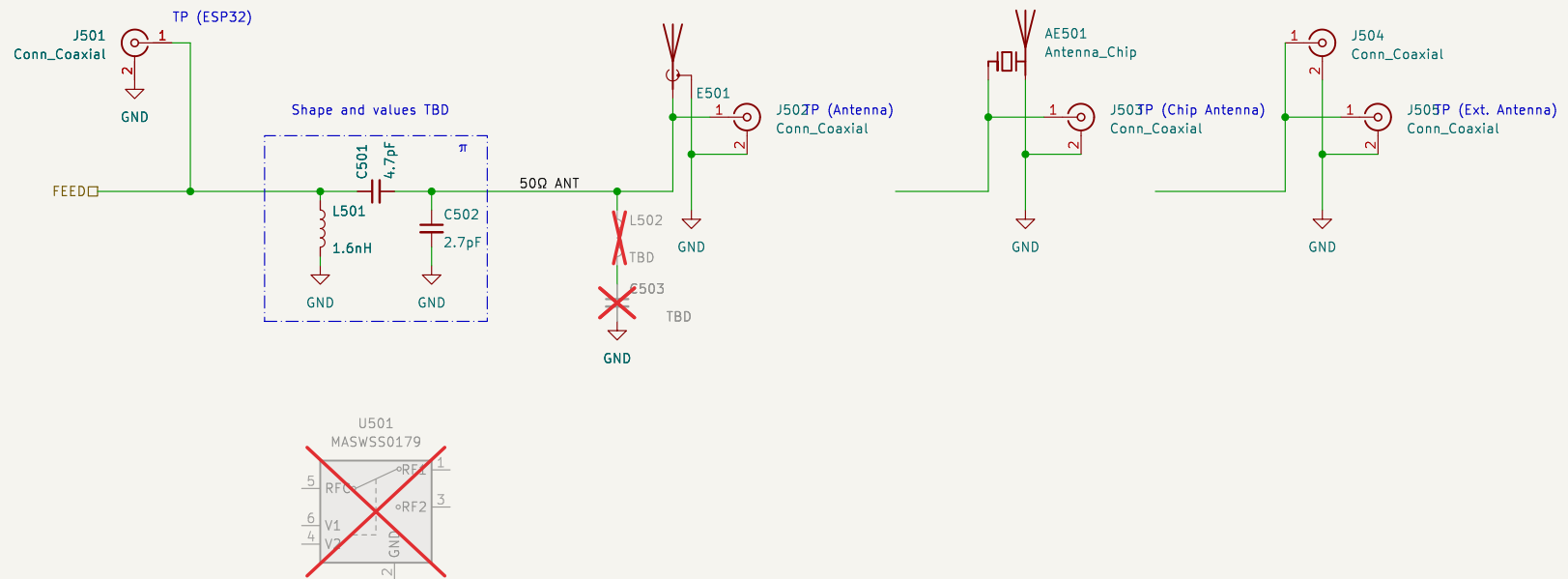
Title: Untitled 4-layer JLCPCB

Size: A4
KiCad E.D.A. 9.0.2

Date: 2024-06-14

Rev: 1.0.0

Id: 9/15



https://datasheet.lcsc.com/lcsc/1912111437_Maxscend-MXD8546FA_C462321.pdf
https://datasheet.lcsc.com/lcsc/2004132010_WILLSEMI-Will-Semicon-WS7808QC-14-TR_C506160.pdf
https://datasheet.lcsc.com/lcsc/2109141830_Shenzhen-Fuman-Elec-FM8625H_C2857391.pdf
https://datasheet.lcsc.com/lcsc/2109141830_Shenzhen-Fuman-Elec-FM1694C_C2857388.pdf
https://datasheet.lcsc.com/lcsc/1804151811_Skyworks-Solutions-SKY13373-460LF_C150853.pdf
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(c) Vincent Dalstra

Sheet: /main/ESP32-S3/ESP32-S3-core/antenna_experimental/
 File: antenna_experimental.kicad_sch

Title: Untitled 4-layer JLCPCB

Size: A4	Date: 2024-06-14	Rev: 1.0.0
KiCad E.D.A. 9.0.2		Id: 5/15